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# FMCW Radar Node for Surveillance Operations


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# Objective

Fig 1: Visual Indicators of High Stress Vs Neutral Intent

- To develop a standalone FMCW radar node for real-time detection and tracking of human targets using mmWave sensing technology.
- To implement onboard motion analytics and threat assessment based on target behavior and movement characteristics.
- To design an autonomous surveillance system with embedded processing and real-time WiFi-based monitoring.

# Introduction

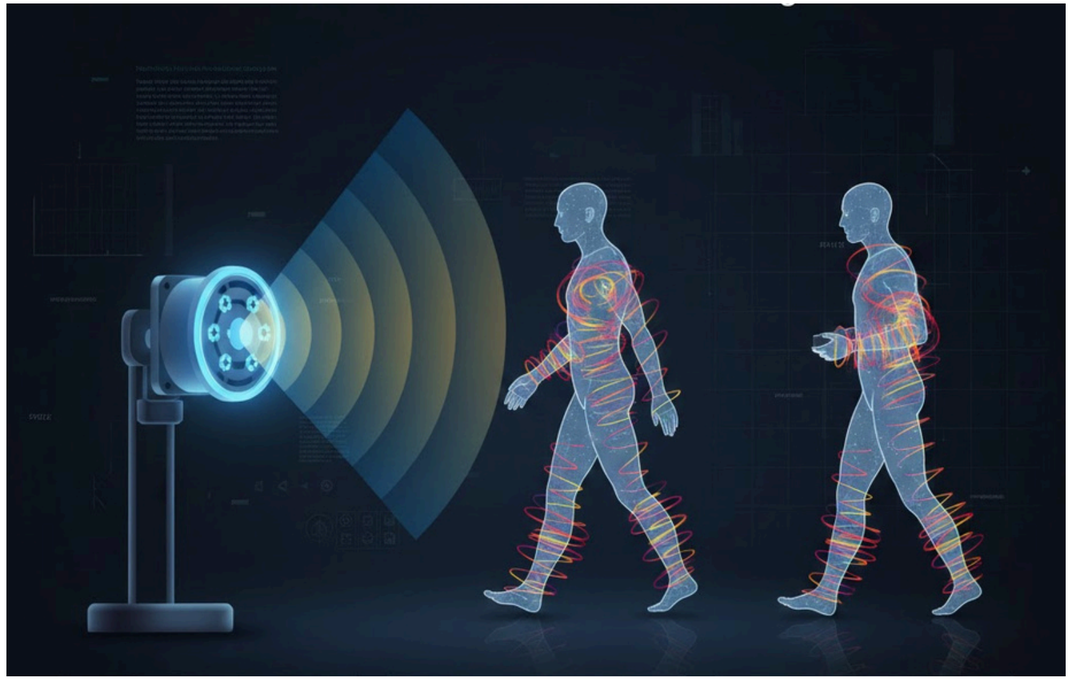


Fig 2 : Radar Intelligence Sensing

- Modern surveillance requires reliable detection that operates independently of lighting conditions and visual obstructions.
- Millimeter-wave radar provides privacy-preserving, real-time detection of human presence even in low-visibility environments.
- This project develops a compact standalone radar node capable of multi-target tracking, behavioral analysis, and real-time threat evaluation.

# Literature Survey

| Reference No. | Article Details | Journal Name | Year of publication | Observations |
|---|---|---|---|---|
| [1] | Multisensor Low-Cost System for Real Time Human Detection and Remote Respiration Monitoring | IEEE | 2019 | A pretrained neural network (MobileNet) is used to process camera frames and detect human presence in real-time. |
| [2] | A High-Computational Efficiency Human Detection and Flow Estimation Method Based on TOF Measurements | Sensors (MDPI) | 2019 | The proposed algorithm mainly consists of head detection based on local pooling and searching, classification refinement based on human morphological features. |
| [3] | Real-Time Human Detection and Gesture Recognition for On-Board UAV Rescue (Liu, C. and Szirányi, T. ) | Sensors (MDPI) | 2021 | UAV rescue system using hierarchical gesture recognition (DNN/CNN). Key: dynamic "Attention" (set) and "Cancel" (reset) gestures. |

# Literature Survey

| Reference No. | Article Details | Journal Name | Year of publication | Observations |
|---|---|---|---|---|
| [4] | A Human-Detection Method Based on YOLOv5 and Transfer Learning Using Thermal Image Data from UAV Perspective for Surveillance System (Mantau, A. J. et al. ) | Drones (MDPI) | 2022 | Optimizes YOLOv5s with Transfer Learning for small-object human detection using RGBT (Thermal + RGB) images from a UAV. Achieved up to 88.8% AP on TIR images. |
| [5] | Using Deep Learning with Thermal Imaging for Human Detection in Heavy Smoke Scenarios | Sensors (MDPI) | 2022 | A thermal imaging camera (TIC) with a deep learning model is used as an intelligent human detection approach during emergency evacuations. |
| [6] | MILLIMETER WAVE RADAR-BASED HUMAN ACTIVITY RECOGNITION FOR HEALTHCARE MONITORING ROBOT (Gu, Z. et al.) | arXiv | 2024 | Proposes RobHAR for real-time, continuous HAR on a movable robot. Uses lightweight LPN-BiLiLSTM model with HMM-CTC for robust transition optimization. |

# Literature Survey

| Reference No. | Article Details | Journal Name | Year of publication | Observations |
|---|---|---|---|---|
| [7] | Radar-Based Human Activity Recognition: A Study on Cross-Environment Robustness (El Hail, R. et al.) | Electronics (MDPI) | 2025 | Solves HAR domain shift in unseen environments. Combines time-frequency warping (DA) and DANN (UDA) to reduce performance drop from 29.6% to 13.2%. |
| [8] | Through-the-Wall Radar Human Activity Recognition WITHOUT Using Neural Networks (Gao, W.) | arXiv | 2025 | Non-NN TWR HAR. Segments micro-Doppler using ACM. Classifies using Mapper algorithm (topological similarity). Showed comparable robustness to noise as NNs. |

# BLOCK DIAGRAM

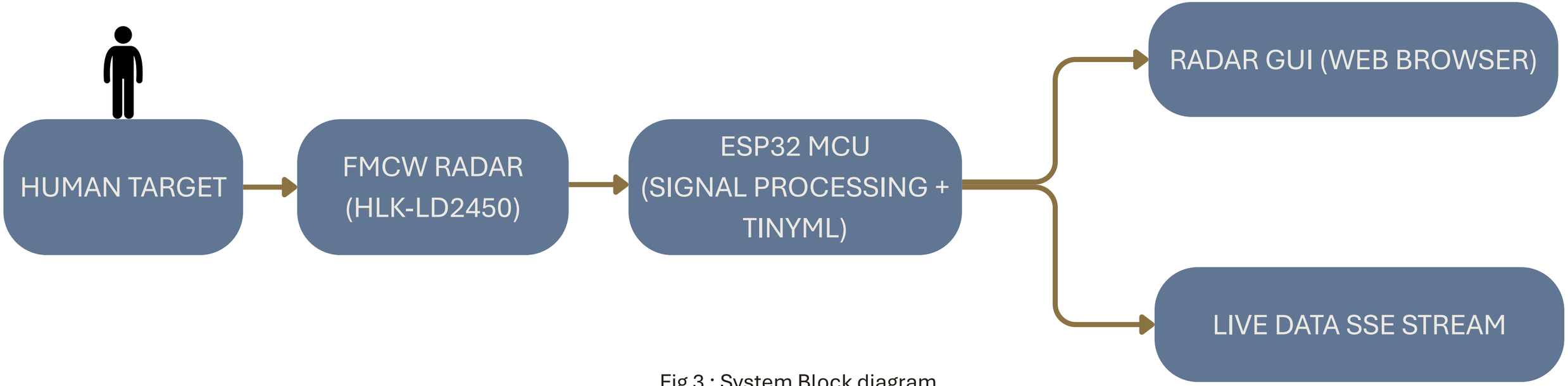



Fig 3 : System Block diagram

# Methodology

- Hardware Design

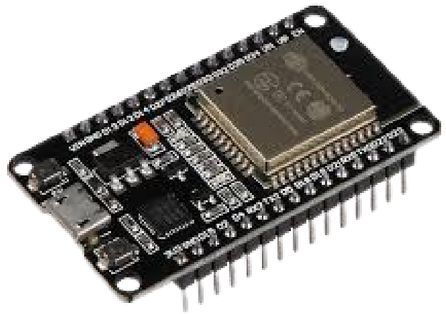


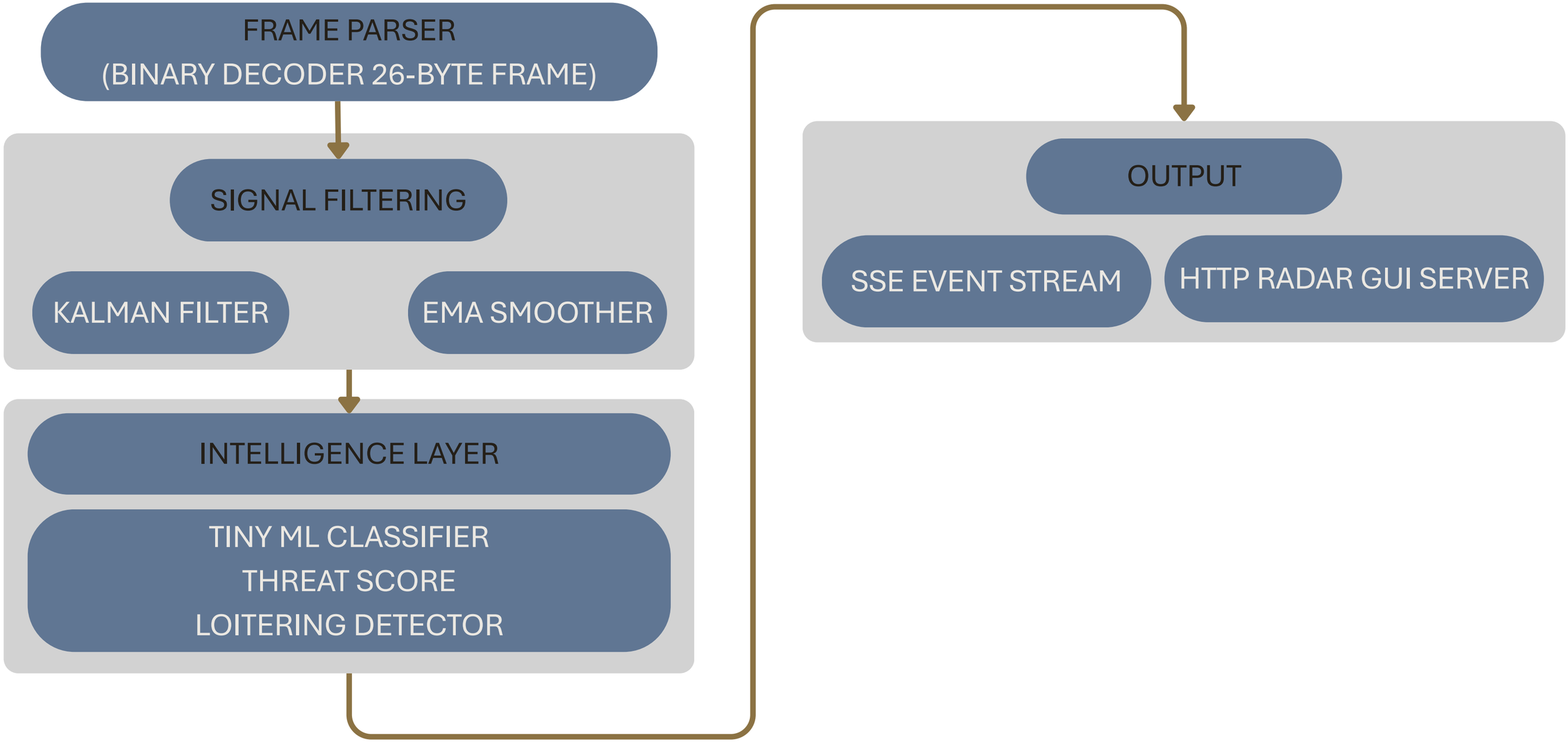
Fig 4 : ESP 32[10]

- Central processing unit
- Executes analytics algorithms
- Hosts embedded web dashboard
- Provides WiFi connectivity

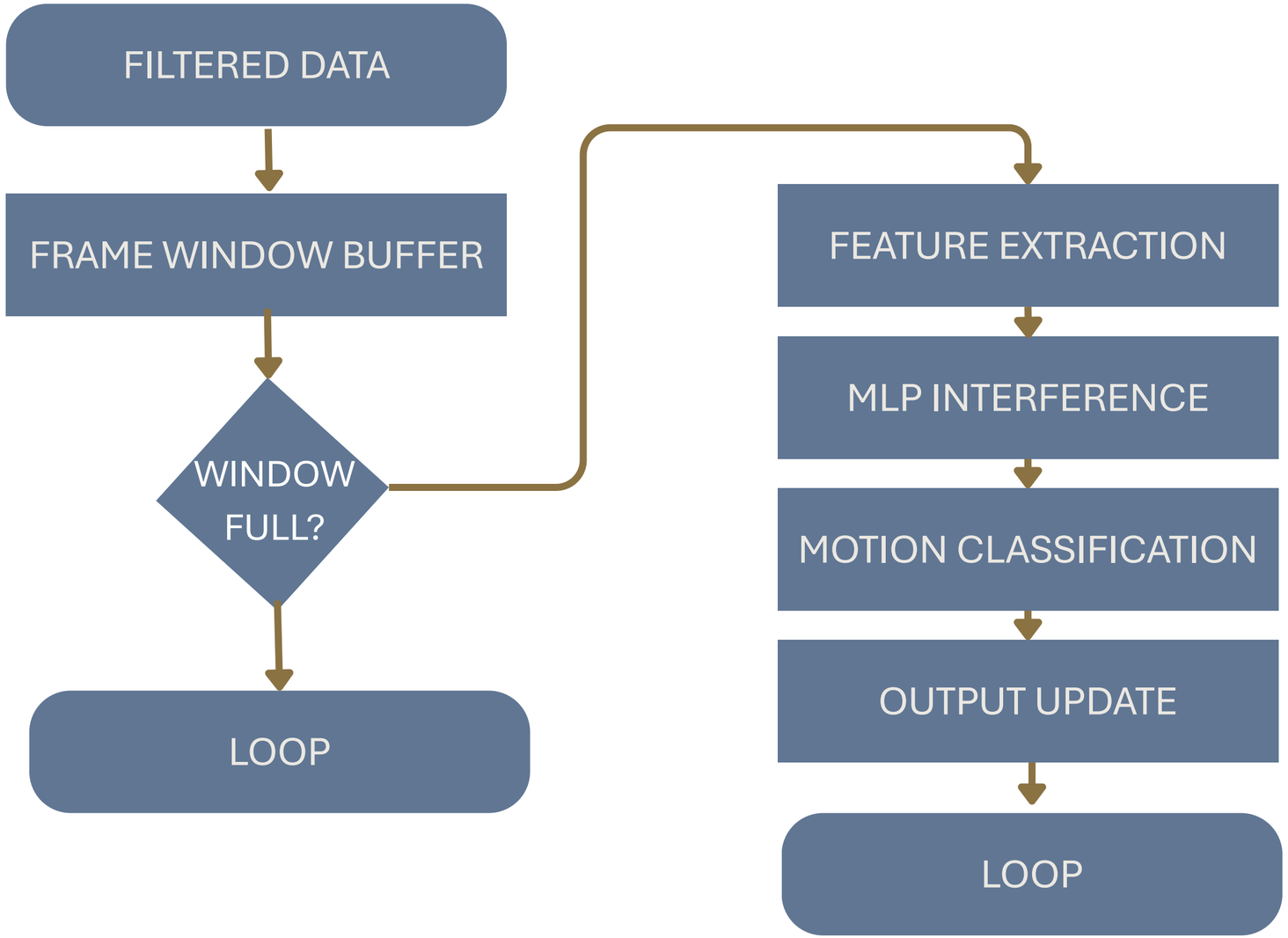
Fig 5 : HLK-LD2450[10]

- 24GHz FMCW radar sensor
- Tracks up to 3 targets simultaneously
- Provides position and velocity data
- Supports zone-based detection

# ESP32 FIRMWARE ARCHITECTURE



# TINYML MOTION CLASSIFICATION PIPELINE



# CIRCUIT DIAGRAM

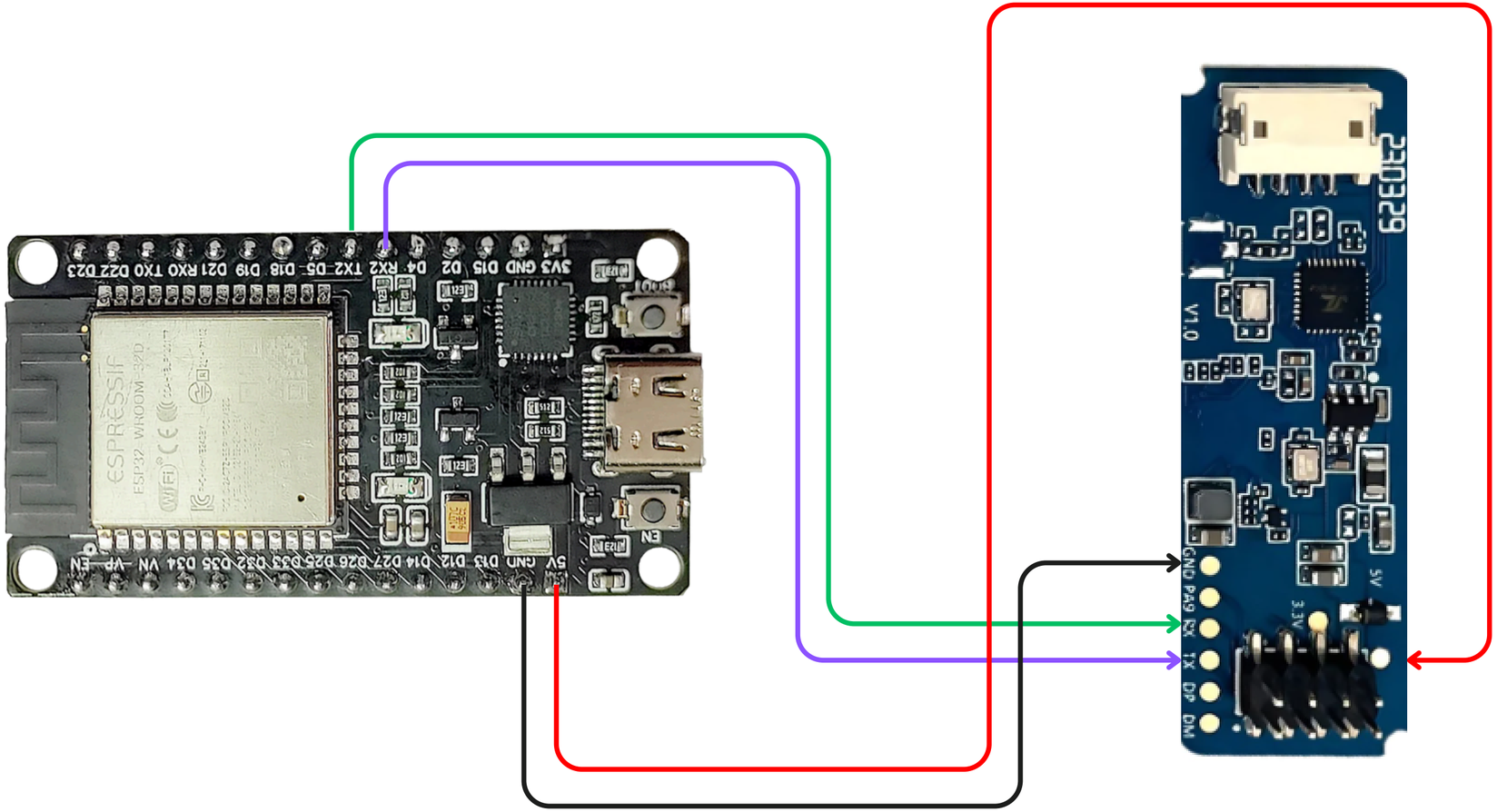



Fig 7 : Circuit Diagram

# EXPERIMENTAL SETUP

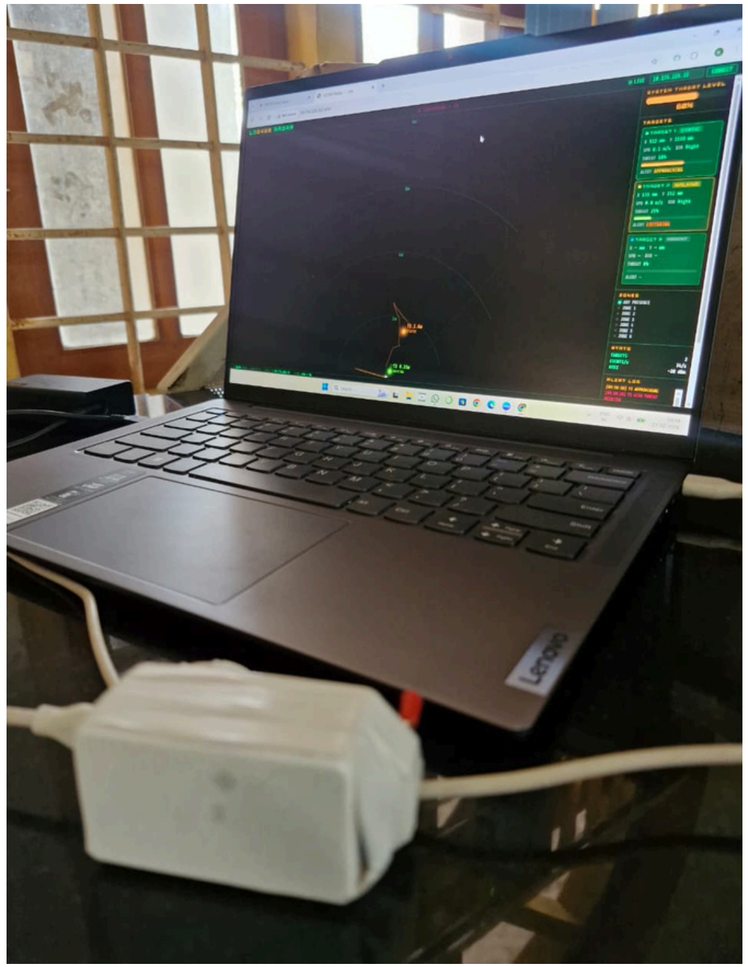



Fig 8 : Experimental Setup

# RESULTS AND ANALYSIS

## Kalman Filter Noise Reduction

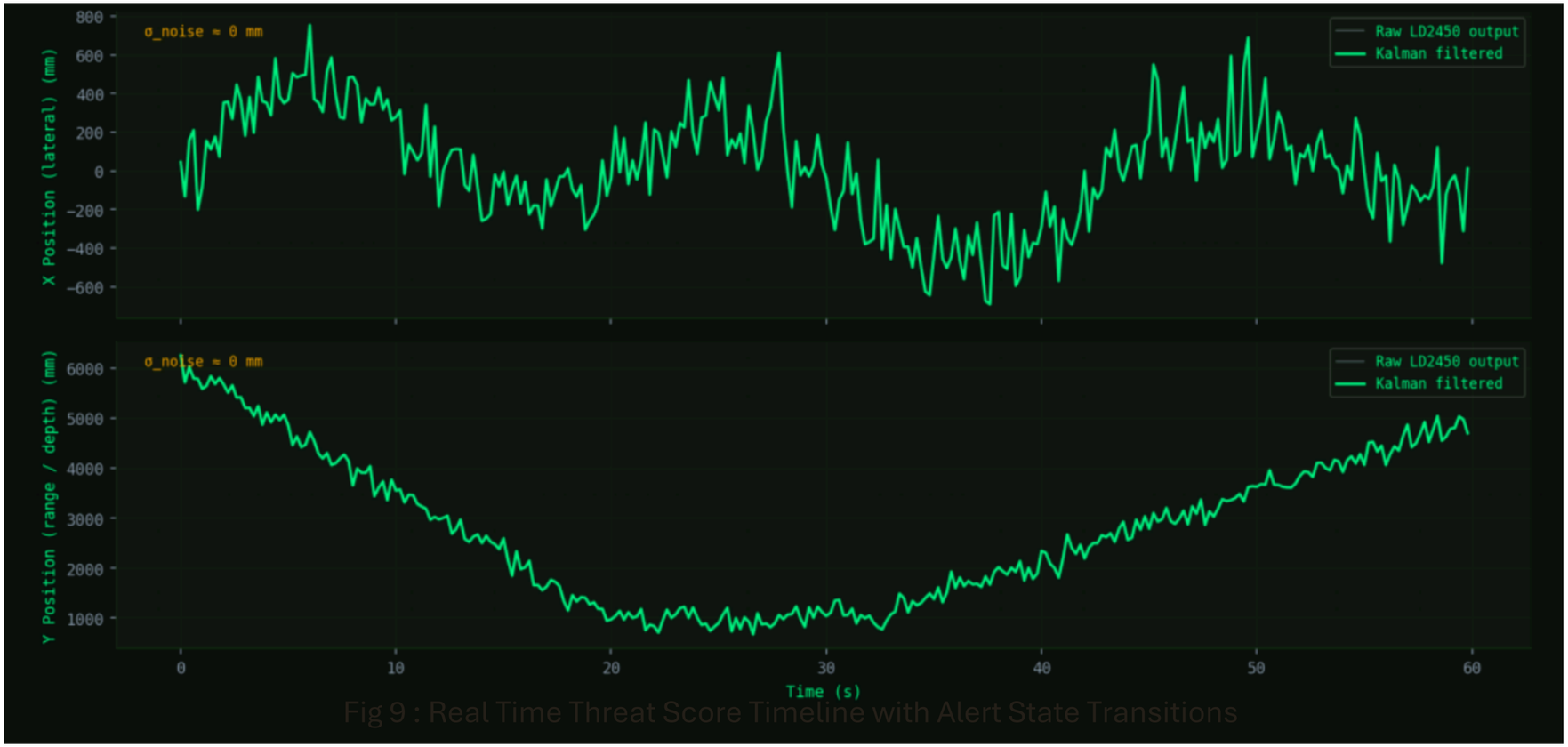



Fig 9 : Kalman Filter Noise Reduction on LD2450 Position Data

# RESULTS AND ANALYSIS

## Real Time Threat Score Timeline

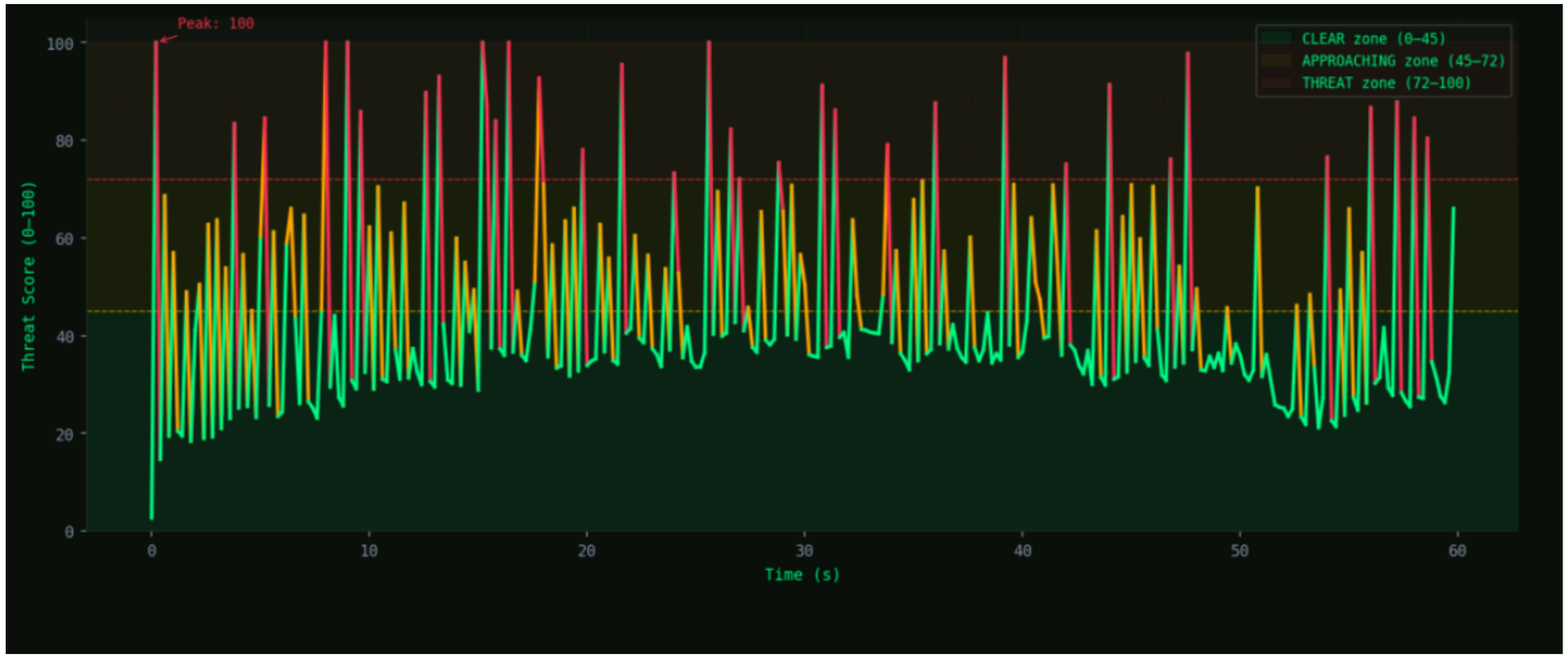



Fig 10 : Real Time Threat Score Timeline with Alert State Transitions

# 2D TARGET POSITION TRACE

Scatter plot of raw vs Kalman-filtered positions in mm. Color-coded by threat score; blue arrows indicate predicted movement direction.

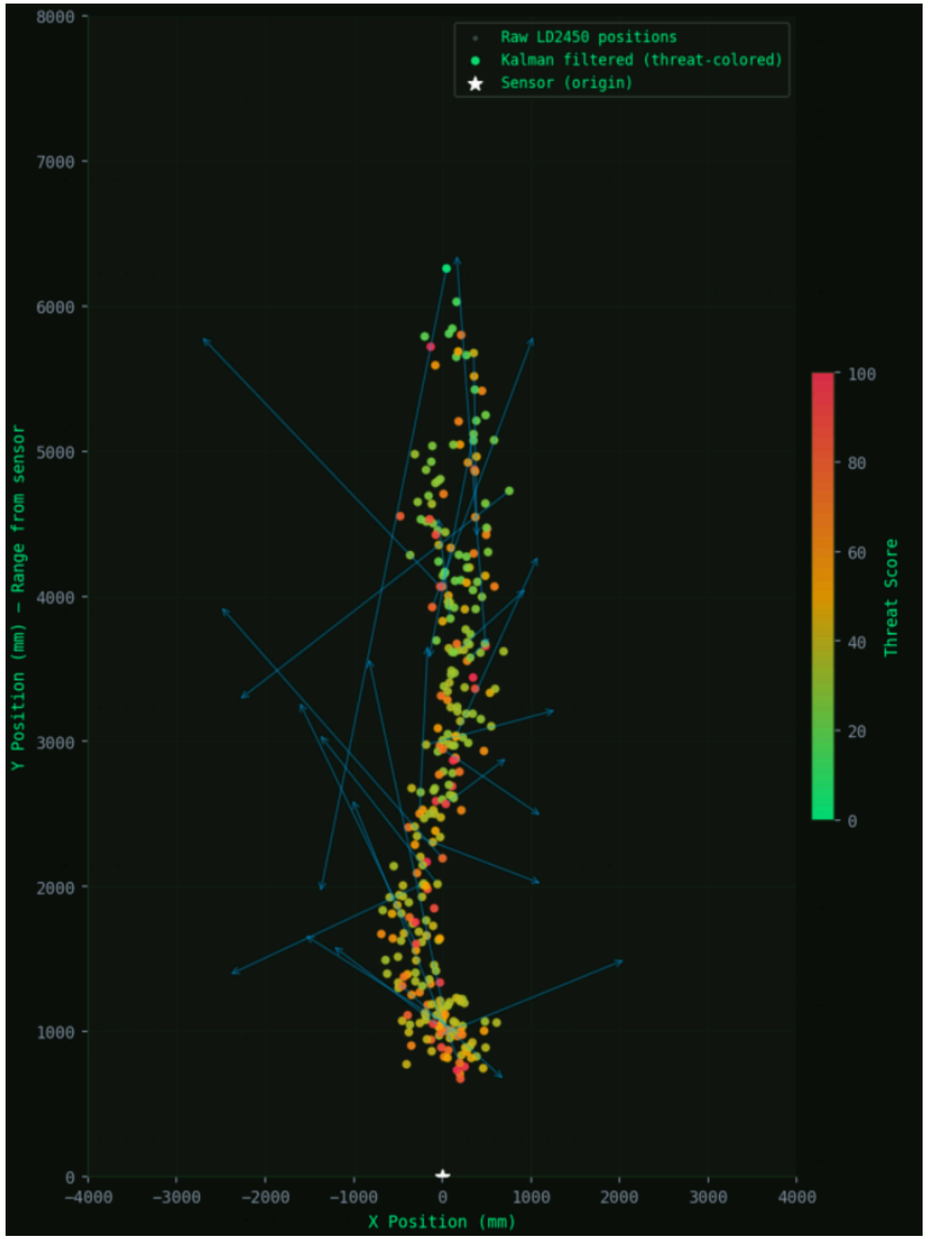



Fig 11 : 2D Target Position Trace(Kalman Filtered+Prediction Vectors)

# MOTION CLASSIFICATION DISTRIBUTION

Fig 12 : Target Classification Results

# SYSTEM COMPARISON CHART

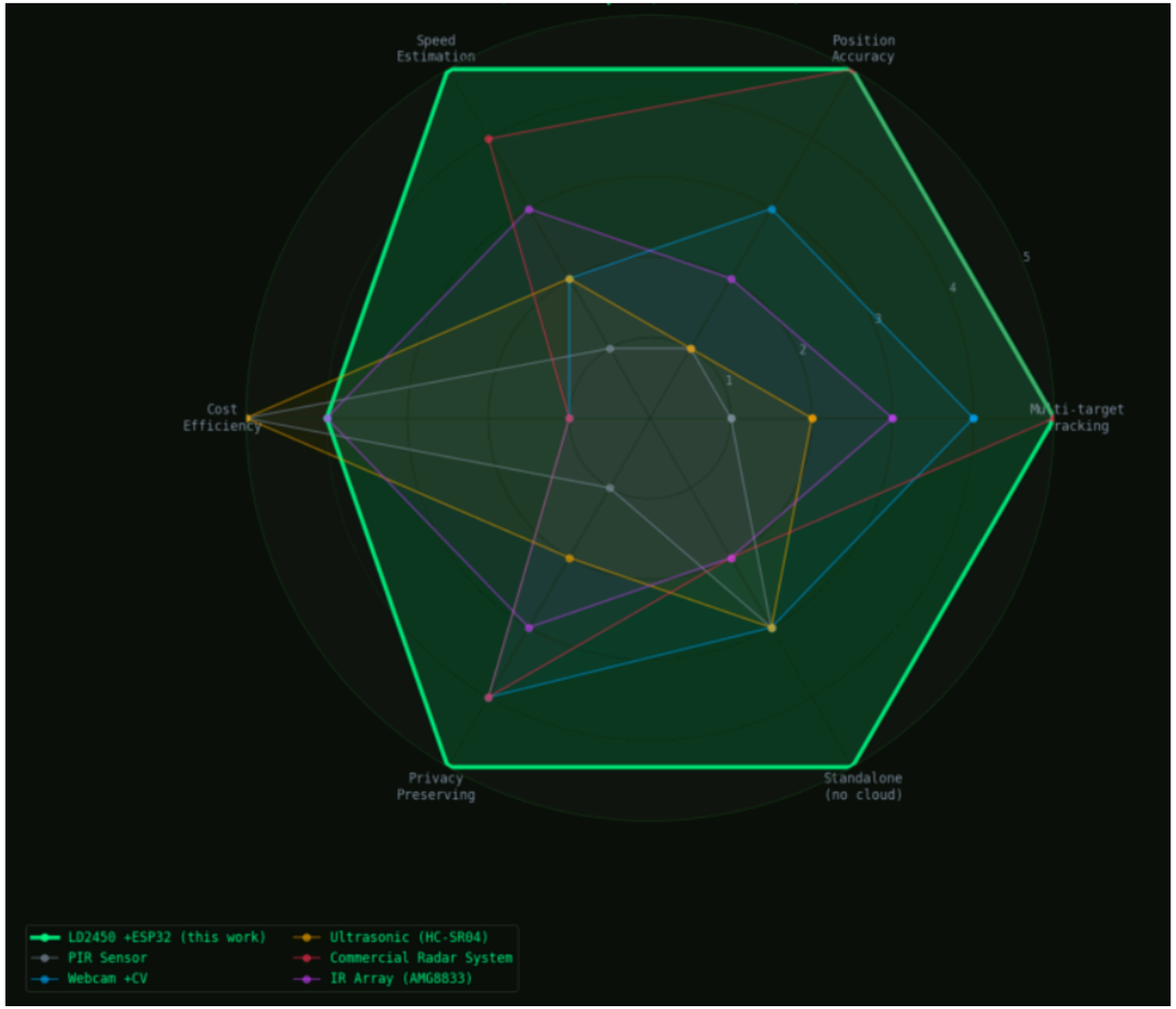



Fig 13 :LD2450+ESP32 Vs Alternatives

# Project TimeLine (Gantt Chart)

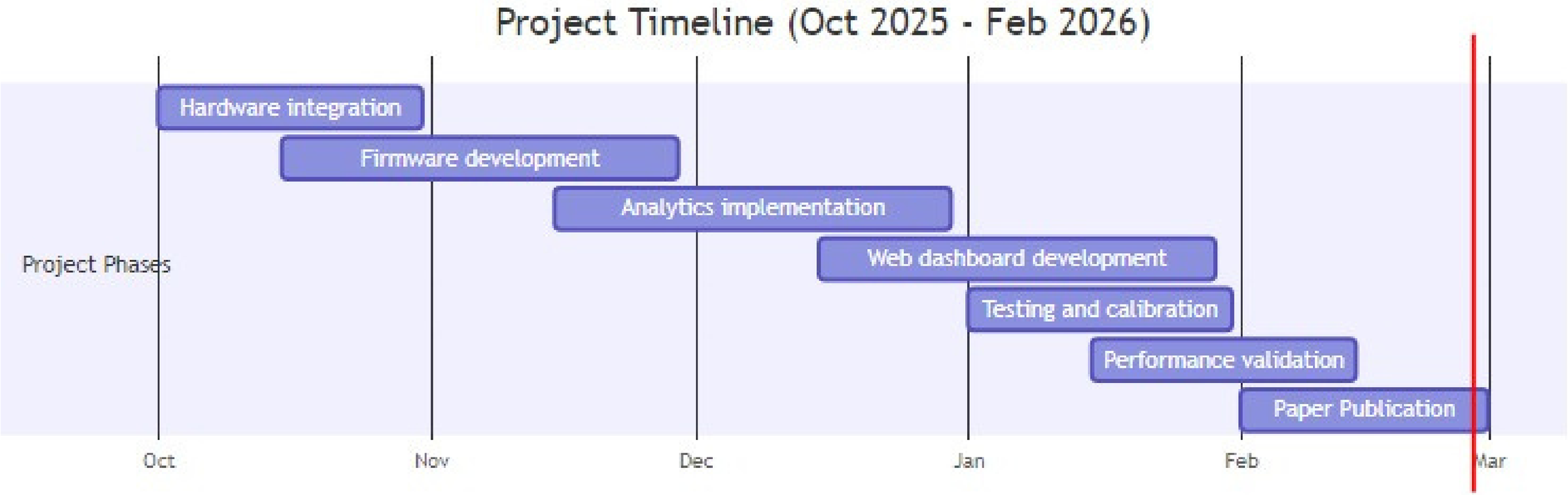



Fig 14 : Gantt chart

# Project Budget

| SlNo | Product Name | Quantity | Price |
|---|---|---|---|
| 1 | HLK-LD2450 Radar | 1 | Rs. 1640 |
| 2 | ESP32 | 1 | Rs. 360 |
| 3 | MISC. | - | Rs. 500 |
| - | TOTAL | 2 | Rs. 2500 |

# Individual Contribution

1. Project Prototype and ML Training - Arjun M
2. Conference Paper Content and ML , Project Report, Presentation - Kripa Krishna
3. Hardware and Firmware Testing - Blesson Binu

# Reference


[1] Ivanovs, Arturs, et al. "Multisensor low-cost system for real time human detection and remote respiration monitoring." 2019 Third IEEE international conference on robotic computing (IRC). *IEEE*, 2019.

[2] Wang, Weihang, et al. "A high-computational efficiency human detection and flow estimation method based on tof measurements." *Sensors* 19.3 (2019): 729.

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[5] Tsai, Pei-Fen, Chia-Hung Liao, and Shyan-Ming Yuan. "Using deep learning with thermal imaging for human detection in heavy smoke scenarios." *Sensors* 22.14 (2022): 5351.

[6] Gu, Zhanzhong, et al. "Millimeter wave radar-based human activity recognition for healthcare monitoring robot." *arXiv preprint arXiv*:2405.01882 (2024).

[7] El Hail, Reda, et al. "Radar-Based Human Activity Recognition: A Study on Cross-Environment Robustness." *Electronics* 14.5 (2025): 875.

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[9] Google. (2025, Dec. 30). "Visual Indicators of High Stress Vs Neutral Intent." [AI-Generated Image]. Generated by Gemini.

[10] "Comparative Study of Microcontroller: ARDUINO UNO, RASPBERRY PI 4, ESP 32" (2024). International Journal for Research in Applied Science and Engineering Technology (IJRASET).